Network Analysis 2026

Group Project Activity

Description. The group activity aims to allow students to work and learn collaboratively while developing a project that leverages Network Science methods to seek insights into research questions. Projects can focus on the empirical or theoretical insights. It is an open project; students can suggest specific topics and possible data sources or approaches.

Format. In groups of up to four students, but recommended to be four, you can use any data source and the tools of your preference (e.g., programming language or software).

Delivery.

- Five-page report
- Deck of Slides for a 15-minute presentation.
- Template will be shared on Moodle.

Important Dates (two deliveries).

- Project Proposal Presentation April 9th
- Project Delivery and Presentation in May/June (date to be defined **TBA**)

Below are some suggestions for your assignment/project.
Please feel free to share your ideas with us.

First Delivery

- Present 5 slides in no more than 5 minutes
- Introduce your network, discussing its nodes (N) and links (L)
- Tell us how you will collect the data and provide an estimate of the network size (N, L). Make sure N > 100
- Tell us which questions you plan to ask.
- We understand they may change as you advance with your project and the class
- Tell us why you care about your network.

Final Delivery

- Each group has 10 minutes to present their final project.
- The presentation will take place in the last week of classes.
- Tell us about your data collection method and show us a record of the source data to give a sense of where you started.
- Measure: N, L, and their time dependence if you have a time-dependent network; degree distribution; average path length; clustering coefficient. (If you have a weighted network, show us the weighted distribution. Depending on what is appropriate for your project, you can: explore community structure, discuss network robustness and/or spreading/diffusion dynamics, etc. Your project can also stem from a scientific manuscript.
- It is not sufficient to simply measure things – you need to discuss the insights you gained, always asking:
    - What was your expectation?
    - What is the proper random reference?
    - How do the results compare to your expectations?
    - What did you learn from each quantity?
- Grading Criteria:
    - Use of network tools (completeness and correctness)
    - Ability to extract insights and properly discuss the outcomes
    - Overall quality of the project/presentation
- No need to write a report; however, you will have to deliver the slides ahead of the presentation

Analysis of Existing Datasets

The goal is to analyze a network dataset using the concepts you learn in the UC. The topic is open. You should explore one that interests you.

You might choose a small or large graph. Depending on the domain, you might need to use different tools, address different obstacles, and answer related questions. **Look for articles associated with the dataset of your choice and start by repeating and extending their analysis.** Often, these articles include the characterization of the network to evaluate the importance (centrality) of each node using relevant centrality measures for the chosen dataset and problem or to suggest a set of principles responsible for the self-organization and creation of such network topology/topologies.

Examples of topics:

- Online social media platforms offer an excellent source of network data. User relationships often reflect positive (friendly) and negative (antagonistic) interactions, translated into networks where edges have an associated sign. Thus, you can use this platform to dive into theories of signed networks from social psychology, such as social balance (Leskovec et al., 2010b), or to study link prediction in these contexts (Leskovec et al., 2010a).
- Study the topology or inter-community interactions on Reddit, examining cases where users of one community are mobilized by negative sentiment to comment in another community. For an interesting reference, please see Kumar et al. (2018) and the associated dataset available in *Stanford's' Large Network Data Collection* (SNAP).
- Study the competition structure between firms in organized markets by extracting networks from public procurement data. See Lyra et al. (2021) for an example in Brazil and Frederic Sturm et al. (2024) or Curado et al. (2021) for an example using data public procurement data from Portugal.
- Follow the footsteps of Evelina Gabasova[1] and analyze the Star Wars social network or a character network from your favorite movie or book series.
- Natural language is an evolving system whose present structure can doubtlessly be a product of the long history of adaptation and self-organization. Many attempts to study language from the point of view of network science express words as nodes and their relations by edges. If you want to know more about natural languages, you're invited to explore the differences between network characteristics of the texts belonging to different novels/languages/fields (Caldeira et al., 2006; Fellbaum, 2010; Grabska-Gradzińska et al., 2012; Morato et al., 2004).
- Take the flights at https://openflights.org/data.html and explore airport importance, country importance, communities, etc.
- Instead of analyzing an existing dataset, you may dedicate your project to contributing to the community by creating a novel dataset. Please let us know if you have an idea.

Finding communities in graphs is a classic problem for network science researchers and data scientists. A suitable large-scale (unsupervised) clustering method is needed. As a result, community-finding algorithms have been studied by mathematicians, computer scientists, physicists, and sociologists, among others.

[1] http://evelinag.com/blog/2015/12-15-star-wars-social-network/

- Implement and compare two or three algorithms to find communities in different graphs.

For many physical networks, removing nodes can have a much more devastating consequence than the loss of vertices and links whenever the intrinsic flows and maximum loads are taken into account (e.g., think about the power transmission grid, airport networks, etc.). Here, you're invited to model this process and engineer some control measures capable of halting a cascade.

- Robustness of a Network of Networks. Network research has been focused on studying the properties of a single isolated network. Here, you test the analytical framework proposed in Refs through numerical simulations to understand the resilience *n* interdependent networks under random and targeted attacks (Dong et al., 2013; Gao et al., 2011).
- How can we halt cascading failures? In line with the previous project, you shall investigate an efficient defense strategy based on the selective removal of nodes and edges right after the initial attack or failure. The project involves creating a computer model of load balance (Motter, 2004).

The properties of real-world networks profoundly impact dynamical processes occurring in various systems. The study of epidemic spreading is one of the most evident examples. You are invited to explore several simulation and mathematical approaches related to network epidemics in such a line. See Pastor-Satorras et al. (2015) for a general overview of this topic.

- Implement a computer simulation of the SIS model in a given network. Compute the epidemic threshold for lattices, random graphs, BA model networks and minimal model. Discuss the results. Suggested reading: (Pastor-Satorras et al., 2015; Pastor-Satorras & Vespignani, 2001), see also (Barabási, 2012; Barrat et al., 2008).
- Try implementing a simple computer model of the co-evolution of disease states (e.g., SIS) and network structure. Assume that, at each time step, individuals connected to infected nodes will try to rewire their links. In other words, network evolution is a natural outcome of information disease states in each neighborhood. Try an agent-based version of the minimal model discussed in Van Segbroeck et al. (2010).

However, other related dynamic processes have also intrigued the community. For instance, why is it so complex to eradicate social segregation? In 1971, the American economist Thomas Schelling created an agent-based model that might help explain why segregation is so difficult to combat. His segregation model showed that even when individuals (or "agents") didn't mind being surrounded or living by agents of a different race, they would still choose to segregate themselves from other agents over time! Although the model is quite simple, it gives a fascinating look at how individuals might self-segregate, even when they have no explicit desire. The Schelling model of segregation nicely illustrates how individual incentives and individual perceptions of difference can lead collectively to segregation.

- What if, on top of Schelling's original assumptions, we allow agents to adapt their tolerance to others in response to their local environment? Let's say that when agents are exposed to the out-group, their tolerance increases if they are currently satisfied with their environment, but otherwise, it decreases. Does adaptive tolerance increase segregation? Try to answer this question following Urselmans & Phelps (2018). In this project, you are expected to create a computer simulation and analyze Schelling's model. See Easley et al. (2010) and Schelling (1971) for more details.

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